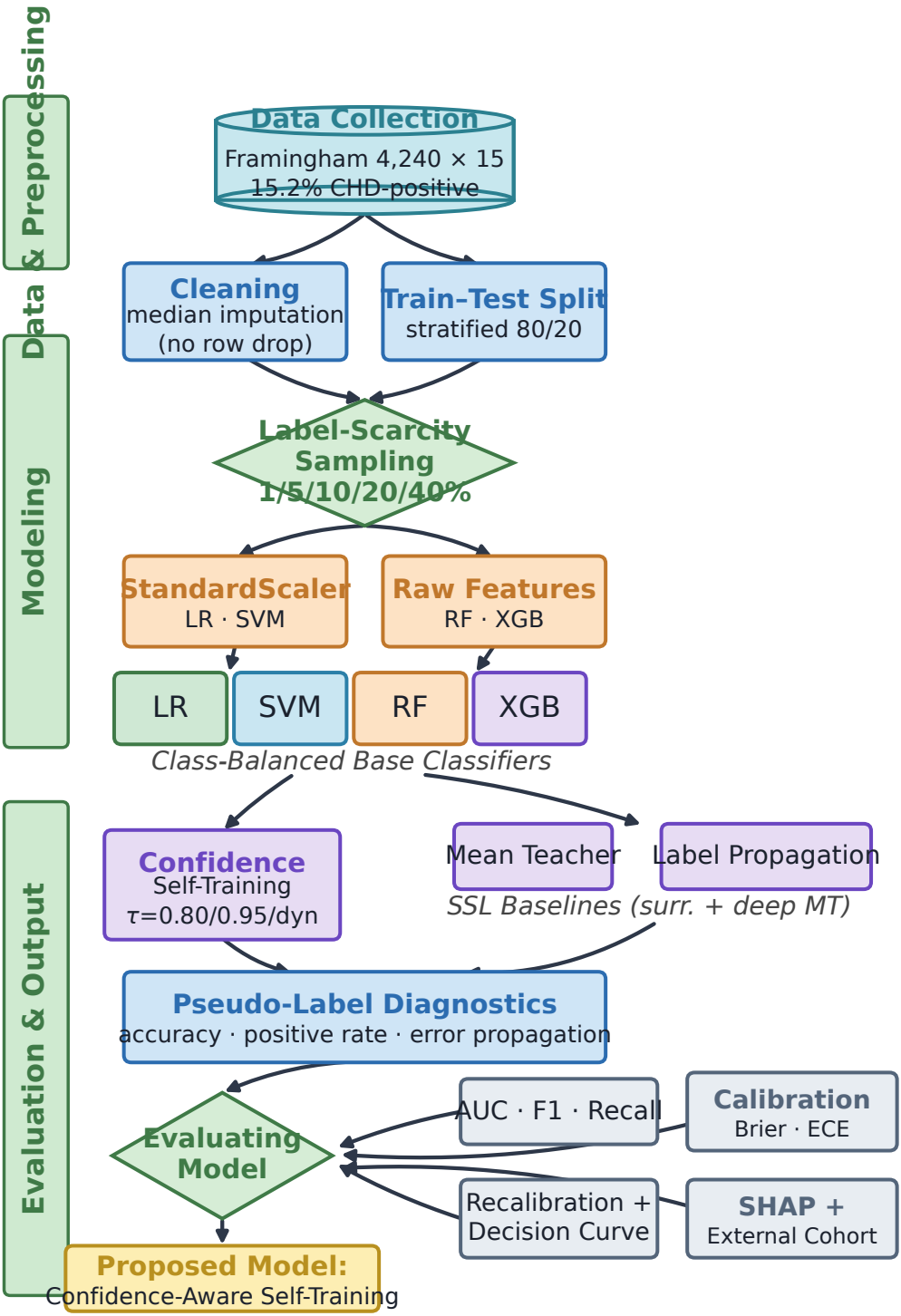
Data & Preprocessing
Modeling
Evaluation & Output
Data Collection
Framingham 4,240 × 15
15.2% CHD-positive
Cleaning
median imputation
(no row drop)
Train–Test Split
stratified 80/20
Label-Scarcity Sampling
1/5/10/20/40%
StandardScaler
LR · SVM
Raw Features
RF · XGB
LR
SVM
RF
XGB
Class-Balanced Base Classifiers
Confidence
Self-Training
τ=0.80/0.95/dyn
Mean Teacher
Label Propagation
SSL Baselines (surr. + deep MT)
Pseudo-Label Diagnostics
accuracy · positive rate · error propagation
Evaluating Model
AUC · F1 · Recall
Calibration
Brier · ECE
Recalibration + Decision Curve
SHAP +
External Cohort
Proposed Model:
Confidence-Aware Self-Training